

PORTFOLIO · 2026

Muhammad Mahameru Abidin

Data Scientist | AI Engineer

Building end-to-end AI solutions — from research and model development to scalable production systems. This portfolio highlights real-world work across Computer Vision, Machine Learning, Generative AI and Full-Stack Web development, designed to turn complex data into measurable business value.

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Location Sidoarjo, East Java, Indonesia

13+

Projects

4

AI & Web domains

~2 yrs

Experience

Computer Vision

Real-time visual intelligence from camera feeds.

PPE Detection System

[Computer Vision](#)

An AI computer-vision system that detects personal protective equipment — hardhats, safety vests and more — on workers in real time, turning existing cameras into an automated workplace-safety compliance monitor.

YOLO Python OpenCV Flask Google Cloud Docker

Face Mask Detection System

[Computer Vision](#)

Real-time face-mask compliance detection across crowded public and workplace environments, classifying masked/unmasked faces with confidence scoring to support health & safety policies.

YOLO Python OpenCV Flask Google Cloud Docker

Face Recognition System

[Computer Vision](#)

Real-time face recognition that identifies and verifies individuals from camera feeds against a registered database — built for attendance, access control and security use cases.

Python OpenCV TensorFlow

Machine Learning

Predictive models and analytics for real-world decisions.

Smart Sootblow Rule Analysis

[Machine Learning](#)

ML system for power-plant sootblowing optimization. Analyzes existing operating rules and recommends improved, data-driven ones — building and comparing Linear Regression, Polynomial and Multivariate models (scored by R^2 and eligibility) on real plant sensor data.

Python Scikit-learn Jupyter MySQL Flask Swagger Docker

Anomaly Detection System

[Machine Learning](#)

Monitors sensor and operational data to detect anomalies in real time, flagging unusual patterns that may indicate faults or risks — aimed at industrial monitoring where early detection prevents downtime.

Python Scikit-learn FastAPI Docker

Time-Series Forecasting

[Machine Learning](#)

Forecasting system that predicts future values from historical time-series data — such as demand, load or sensor readings — to support planning and data-driven decision-making.

Python Scikit-learn Pandas

ILO/KBLI Code Recommendation

[Machine Learning](#)

Automated system that crawls and scrapes web and social-media data to understand a company's activities, then recommends the most relevant ILO/KBLI business-classification codes — replacing slow, manual classification.

Python Pandas FastAPI

Generative AI

LLM-powered, document-grounded assistants.

XORCA Enterprise AI Assistant

[Generative AI](#)

Enterprise AI assistant powered by Claude — a document-grounded chatbot that answers questions about a company's products, capabilities and industries. Served through a Flask API with Redis caching and deployed on Google Cloud.

Python

Flask

Swagger

Redis

Google Colab

Google Cloud

Claude API

Full-Stack Web

End-to-end web systems — from database to user interface.

LPPM Research & Community Service System

[Full-Stack Web](#)

Multi-role enterprise platform digitizing a university research office (LPPM): lecturers submit and report research/community-service proposals; admins manage master data (SDGs, SKIM, funding, labs, outputs) via role-based dashboards.

Laravel Blade Bootstrap PostgreSQL

Coffee Shop POS (Cashier) System

[Full-Stack Web](#)

Point-of-sale cashier application for a coffee shop — transactions, menu, members and points, vouchers, tax and receipt printing, plus an admin area and sales reporting for owners.

Laravel Blade Bootstrap MySQL

E-Commerce Website

[Full-Stack Web](#)

Full-stack e-commerce platform covering the complete online-shopping journey — product browsing, search, shopping cart, user accounts and checkout — built end to end from database to a responsive storefront.

Laravel MySQL PHP Vue.js Bootstrap

Library Management System

[Full-Stack Web](#)

Library management system handling members, book collections and borrowing transactions, with an admin dashboard that digitizes day-to-day library operations and replaces manual record-keeping.

MySQL PHP CSS3 JavaScript

Kominfo Activity Information Portal

[Full-Stack Web](#)

Information portal for publishing and managing Kominfo (BPSDMP) activity news — public article pages, search and comments with a secured admin area for content management. Built for the BNSP Junior Web Developer certification.

PHP MySQL Bootstrap JavaScript